AN ECLECTIC TECHNOLOGIES FOUNDATIONAL DEFINITION

FD — Foundational Definitions

November 2026 Edition — Contents and Edition Statement

Edition of 19 October 2026

Tim Roxey

Eclectic Technologies

tim.roxey@gmail.com · 410.474.9240 (mobile/Signal)

SUGGESTED CITATION

T. Roxey, ed., *FD — Foundational Definitions*, *November 2026 Edition*, Eclectic Technologies, October 2026. https://doi.org/10.5281/zenodo.23401272

Cite this edition: https://doi.org/10.5281/zenodo.23401272. Cite the family as it currently stands, across all editions: https://doi.org/10.5281/zenodo.22181028. The October 2026 Edition remains citable at https://doi.org/10.5281/zenodo.23117612, the September 2026 Edition at https://doi.org/10.5281/zenodo.23000686, and the August 2026 Edition at https://doi.org/10.5281/zenodo.22181029, each for what it contains.

---

1. What this edition is

The Foundational Definitions are the instruments that own the concepts the rest of the corpus applies. A sector framework, a facility-class design basis, a vendor companion or an application guide uses the Bright Line, the Broker, the envelope, the determinism boundary, local sovereignty and the design basis layers. It does not define them. This family does.

A concept belongs in this family when the correctness of other documents depends on agreeing what kind of thing it is. A concept with a settled external meaning, on which no document's correctness turns, does not belong here however important it is.

This is the family's fourth edition. It does two things. It adds an eighth member, FD-DBL, *Design Basis Layers*, which defines the three layers of design basis analysis on which the whole corpus rests: the Design Basis layer, the accident layer and the adversary layer, the service function that holds them together, and the change in what success means at the adversary layer. And it takes into three existing members the answers the author gave, between 8 and 14 October 2026, to questions the sector work put to the family. It supersedes the October 2026 Edition as the current statement of the family and does not withdraw it.

2. Contents

Designation	Title	Version	Class
FD-BL	The Bright Line	0.6	Foundational Definition
FD-LD	Layer Decomposition	0.5	Foundational Definition

	and Determinism		
FD-EV	The Envelope	0.2	Foundational Definition
FD-BR	The Broker	0.11	Foundational Definition
FD-BL-D1	Mode of Discharge: Pre-Authorization by Verified Gatekeeper	0.5	Determination under FD-BL
FD-BL-D2	Indication Integrity: Trustworthy Basis for Authorization	0.5	Determination under FD-BL
FD-LS	Local Sovereignty	0.6	Foundational Definition
FD-DBL	Design Basis Layers	0.16	Foundational Definition
FD-REG	The Governing Stack Register	0.33	Companion register (dated copy)

Determinations are not Foundational Definitions. They settle a question the governing definition left open, under its authority, and are cited alongside it.

What DRAFT means in this family. Every instrument here is below version 1.0, and each carries DRAFT on its cover. In this family DRAFT means that the instrument is issued and citable at the version stated, and that its author does not yet hold it complete. It does not mean unreleased or provisional in force.

Dates. Each instrument's cover and administrative block carry the date of the version printed on it. The edition's date appears in its citation block and nowhere else.

FD-LS is originated by Project 5 of this portfolio, Grid Operations and Energy Policy, which holds the construct. The instrument here is its definition of record, kept with the rest of the family because that is where the corpus holds what every project inherits. Its own §2.3 states the origination and the lineage.

FD-REG is included as a companion and not as a member. It holds the version state of the family, and the bare citation form used throughout the corpus (FD-BL §4.2, without a version) resolves against it. The copy here is dated with the edition and records every member at the version listed above. After that date its master governs.

3. What changed since the October 2026 Edition

Four instruments move and four do not. FD-EV, FD-BR, FD-BL-D1 and FD-LS stand at the versions issued in the October 2026 Edition. Placement remains consequence, action by action, throughout, and no change here alters the placement criterion, the scope rule or the obligation.

FD-DBL, new at 0.16. *Design Basis Layers* states what the corpus had enacted and never defined. The Design Basis layer, the accident layer and the adversary layer are three analyses of one service function, the thing the facility is designed to do, under assumptions that each layer relaxes. The accident layer carries three assumptions, and the adversary layer is reached only when an actor shows all four canonical characteristics: intentionality, persistence, precision and denial. At the Design Basis layer success is delivery of the function within its operating limits, and at the accident layer it is recovery or a safe, known condition reached with external support; at the adversary layer it is that the facility reaches and holds a protected state, with consequences bounded, using only resources present at the facility.

The instrument also carries the operating-limits ladder, the layer as an input to AI governance, and cyber as a thread through every layer rather than a layer of its own. §6.4 records how a real event that falls short of the four characteristics is recorded, and §4.5 states that reaching the military-action tier moves the duty to defend to the state where the law so places it, and never moves the duty to reach and hold the protected state. §4.4 states the adversary's aim as defeating the function or a requirement on how it is performed, such as that it be performed safely, whether by stopping it, seizing the facility or turning its plant or material against others, and §6.4 records that falsifying the operators' view of the process is precision and not denial.

FD-BL, 0.5 to 0.6. §4.3 gains three paragraphs after its closing rules. A vendor's release to a population of hosts outside any one facility is not by itself an action; its vendor-side controls are obligations the asset owner-operator places on the vendor by contract, and taking such a release in at the facility is an action placed on its consequence that enters only through a governed act. Each widening of a staged release is its own action, and the first release is placed on its whole planned reach. A change a component within scope makes to an enterprise network is an action placed on its consequence.

FD-LD, 0.4 to 0.5. §4.3 states that the carve-out does not cover a component while it runs a configuration other than the one formally tested, including one altered outside a governed act, and the open item at §7 widens to take that condition in. The Record gains R.7, tracing the layer split to the author's formal-methods papers of 2025, with two references cited as provenance only.

FD-BL-D2, 0.4 to 0.5. §4.3 gains a sufficiency paragraph: an indication can be correct and insufficient, and what is missing is identified and sought before an authorization is relied on, a basis acted on while incomplete being recorded as incomplete.

4. Precedence and propagation

Foundational Definitions take precedence over sector frameworks, facility-class documents, application guides, vendor companions, annexes, qualification standards and implementation guidance. A restatement may be expressed in sector vocabulary and may apply a definition to sector-specific actions. It shall not vary the placement criterion, the granularity, the scope of application or the obligation.

Propagation is by reference to the owning instrument, not by restating its text. An instrument that rebroadcasts its own wording re-introduces variance at every revision cycle.

5. What a reader needs from outside this family

The family applies a small number of things it does not define. A reader applying it needs them, and this section says where they are. It points and does not restate.

The severity threshold and the consequence-severity categories. FD-BL §4.2 places an action above the line when its incorrect performance could produce a consequence at or above "the severity threshold established for the facility." The six consequence-severity categories, and the criticality bands built on them by taking the most severe, are stated in the Consequence-Severity Tier Model and applied in the Command Broker Framework, Cross-Sector, §3.3. The corpus assigns the band for an application in a given facility to that facility's asset owner-operator, as the DBA-VC Vendor Companion states. This family does not say more than that about who establishes a facility's threshold, and this edition does not settle it.

The qualification of a Broker. FD-BR §4.3 sets competence by the paired Line's placement criterion and makes qualification non-transferable. The competency instruments, onboarding and the advance-mode competence of the Command Broker are carried in the CB-IB Qualification and Onboarding Standard.

The threat assessment. FD-DBL places an actor in the adversary layer by the four characteristics at the level characterized in the facility's threat assessment (FD-DBL §4.4), and §6.2 leaves the adversary envelope to that assessment. The threat assessment is the facility's and is not part of this family.

The citation form. The bare citation form, by designation and section without a version token, rests on the Corpus Rules note on citation form and the FD stack, and resolves against FD-REG.

6. What this edition asserts, and what it does not

An edition is a coherence claim, not merely a package.

This edition asserts that the eight instruments in §2, at the versions listed there, were read against one another on the date of issue, and that every cross-reference between them **resolves**. In this family a reference resolves when the cited section exists in the version cited and carries the content relied on, not merely when a section of that number exists. The read covered applications of each construct, tables included, and not only its statements.

It does not assert that the corpus applying these instruments has been conformed to them. Conformance of sector and application instruments is staged at each instrument's own next revision, and sibling projects whose instruments are affected were sent notes as each change was made. Where a sector document's statement of one of these constructs differs from the instrument here, the sector document is a restatement and the instrument here governs.

Coherence decays, and it decays fastest in the newest work. The edition date is the honest boundary of the claim.

7. Not in this edition

The surveillance branch is not in this edition: FD-SBL, the Surveillance Bright Line, with its determinations FD-SBL-D1 and FD-SBL-D2 and its register FD-SBL-REG. The branch belongs to Project 1 of this portfolio and is held from release pending external review. Two members of this edition depend on it, and the dependence is stated here rather than left for a reader to meet.

FD-BR cites FD-SBL §4.4, §4.5 and §6.1; FD-SBL-D1 §3, §4.3 and §5; FD-SBL-D2 §3.1 to §3.2, §4, §6, §7.1, §7.3 and §8; and FD-SBL-REG. These carry the Influence Broker's competence, its unit of authorization, its substitutability value and the ground of that value, and the permissible-influence boundary. FD-LD cites FD-SBL §4.4, for why the determinism carve-out does not transfer to the surveillance Line, and FD-SBL-REG. None of these references is satisfied within this edition. They will be when the branch is published.

Sector instantiation is not here by design. FD-BL-D1 §7 and FD-BL-D2 §7 are sector-neutral and delegate instantiation to sector annexes, which are separate instruments and are not part of this family.

The formal statement of the corpus's axioms and premises, as distinct from its definitions, is not in this edition. The author has deferred it to the December 2026 Edition.

8. Corpus-rule designators

Some instruments here name the corpus rules under which their own revisions were made. The rules are not part of this family and bind nothing here. They are glossed once so that a reader can follow the reference.

CR-AR, Act and Record: an act on a store and the record of that act are one obligation. CR-CC: a sentence asserting the currency, completeness or membership of a body beneath it is a claim about that body, and stands only if the body was read. CR-CF: an instrument meant to be inherited is cited without a version token, and only where a current register holds its version. CR-CO, Capture Before Overwrite: what is owed to a store about to be made to agree. CR-DP: designation prefixes are allocated corpus-wide, not per series. CR-OT, Ownership Transfer: what is owed when a concept changes owners. CR-PB, Publication Boundary: what a published artifact owes when its master moves. CR-SC, Standing Change: a change in what the corpus holds true about itself. CR-TA, Terminal Artifact Read: a rendered artifact is read through before it leaves the project. CR-VA, Verification Against Applications: a conformance pass searches the applications of a construct, not only its statements.

9. How to cite

Cite the concept DOI for the definitions as they currently stand, and the version DOI to pin this edition. Within a citation, name the instrument and section: FD-BL §4.2, in *FD — Foundational Definitions*, *November 2026 Edition*. The instrument's own version number is printed on its cover and in its citation block, and FD-REG holds it.

10. Revision policy

Instruments revise on their own schedules and carry their own version numbers. A change to any instrument does not by itself publish a new edition. An edition is cut when the author judges that the family has moved enough to warrant a new coherence claim, and each edition is published as a new version under the same concept DOI. The editions therefore record which combinations of versions were coherent together. That is the property a corpus built on propagation by reference most needs and most easily loses.

Revision History

0.2 — 17 October 2026. FD-DBL moves from 0.15 to 0.16 before deposit, on the author's decision of 17 October 2026 to bring into this edition his answers to the three questions Project 6 referred from the *Nuclear Sector Framework* §9.2. FD-REG moves from 0.32 to 0.33 with it. §2 records both versions. §3 names FD-DBL at 0.16 and adds the restated adversary aim of §4.4 and the denial ruling of §6.4. The edition date (19 October 2026) and the version DOI are unchanged.

0.1 — 15 October 2026. First draft of the November 2026 Edition front matter, written from the October 2026 Edition's front matter. §1 and §3 state the edition's two changes: FD-DBL enters as the eighth member, and FD-BL, FD-LD and FD-BL-D2 take in the edits the November cycle queued in FD-REG. §2 adds FD-DBL. §5 points to the facility's threat assessment, which FD-DBL relies on and does not define. §6 counts eight instruments. §7 records the deferral of the axioms to the December 2026 Edition. §4 and §8 to §10 stand as in the October 2026 Edition. The same day the edition date (19 October 2026),

FD-REG v0.32 and the version DOI, 10.5281/zenodo.23401272, reserved on 15 October 2026, are filled for the cut. Issued as the front matter of the November 2026 Edition.